

Ashish Molakalapalli

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EDUCATION

Master of Science in Computer Science

Texas A&M University

Aug 2025 – May 2027

College Station, TX

- GPA: 3.833 / 4.0 | Graduating **May 2027** | Coursework: Machine Learning, NLP Foundations & Tech, Artificial Intelligence, Information Storage & Retrieval, Analysis of Algorithms, Operating Systems
- Research focus: LLM Agents and Evaluation, Retrieval-Augmented Generation, Production AI Systems, Foundation Models

Bachelor of Technology in Computer Science & Engineering

IIIT Kottayam

Dec 2021 – Apr 2025

India

- GPA: 9.34 / 10.0 | Selected Research Scholar, IoT Cloud Research Group – Top 5 of 150+ applicants

INDUSTRY EXPERIENCE

AI for Security Operations Engineer, Intern

ServiceNow

May 2025 – Aug 2025

San Diego, CA

- Architected two production LLM agents for automated monitoring and incident investigation across global security infrastructure, deployed via the Now Assist Skill Kit and backed by Claude Sonnet on AWS Bedrock.
- Designed a 14-field JSON schema with confidence-based tier escalation (items below 0.85 auto-promote severity) and few-shot prompt strategies ensuring reproducible, structured LLM output at production quality.
- Engineered a 14-tool parallel data-collection layer covering MID server health, certificate expiry, VA scanning, and Tenable hygiene; surfaced real critical findings including a MID server offline since March 2026.
- Engineered a real-time alerting service delivering health summaries to Microsoft Teams via webhooks, replacing a manual multi-instance review process and cutting incident response latency from hours to minutes.

AI Engineer

Shipsy

Sep 2024 – Jul 2025

Gurugram, India

- Designed LLM summarization and classification pipelines processing 10K+ support tickets daily for DTDC, integrating Slack, Gmail, and GitHub APIs to automate triage and reduce backlog by 35%.
- Built an enterprise knowledge graph with structured entity linking and RAG over internal documentation, improving query response latency by 40%; won Best Idea Award at Shipsy Hackathon 2025.

Research Intern, Reinforcement Learning

CDAC (Centre for Development of Advanced Computing)

May 2024 – Nov 2024

Remote

- Proposed and implemented a Multi-Agent RL scheduler using PPO on simulated HPC workloads, achieving 15% lower energy consumption and 12% faster job makespan versus baseline schedulers.
- Applied centralized training with decentralized execution (CTDE) and cooperative reward shaping; ran ablation studies isolating the contribution of each design choice against independent baselines.

AI/ML Intern

UST Global

Jun 2024 – Aug 2024

Kochi, India

- Developed SmartChat, an LLM-powered enterprise chatbot with intent recognition and entity extraction achieving 95% accuracy; fine-tuned GPT-4o on domain-specific data, improving factual correctness by 30% over a zero-shot baseline.
- Optimized request routing and introduced query-time caching, reducing average resolution latency by 30% and improving end-to-end accuracy by 25% under sustained load.

RESEARCH EXPERIENCE

Research Assistant, NLP Lab

Texas A&M University

Feb 2026 – Present

College Station, TX

- Constructed a 941-job deep learning runtime benchmark spanning 37 model architectures across 9 GPU configurations, with a 37-feature schema capturing batch size, sequence length, precision, parallelism, and optimizer configuration.
- Extended the benchmark with a 1,884-run LLM inference dataset capturing prefill/decode split timing, TTFT, TPOT, and KV-cache pressure across 22 models and 4 GPU types.
- Trained GBDT, Random Forest, and neural regressors reaching 23.7% MAPE ($R^2 = 0.881$) on training runtime and 8.8% MAPE ($R^2 = 0.982$) on inference latency; systematic evaluation identified log-transform of the target as worth 40–70% relative MAPE improvement.

- Ran out-of-distribution analysis exposing catastrophic cross-architecture failure (CNN to Transformer, 454% MAPE) while per-architecture models held 7–19%, motivating a hybrid analytical-plus-learned approach over a purely learned one.

Graduate Researcher, NLP Lab
Texas A&M University

Sep 2025 – Feb 2026
College Station, TX

- Designed an 8-stage automated data processing pipeline converting engineering schematics to structured JSON graph outputs; integrated YOLOv8 for component detection, PaddleOCR with EasyOCR for text extraction, and HDBSCAN with UMAP for clustering.
- Containerized the pipeline as a REST API service with structured JSON output contracts, enabling integration with downstream reasoning and knowledge graph construction systems.

PROJECTS

Second Thoughts: When LLM Self-Correction Helps vs Hurts | *Llama-3.1-8B, Mistral-7B, Qwen-2.5-7B, LoRA, GSM8K, HumanEval, MBPP*

- Ran a controlled 3-model $\times$ 5-strategy $\times$ 4-dataset evaluation (96 conditions, 71,595 question-strategy pairs); correction broke 14.7% of correct answers while helping only 4.4% – a $3.4\times$ damage ratio – with self-consistency dominating correction in 11 of 12 compute-matched comparisons.
- Showed the same recipe regresses on code (MBPP – 10.27, HumanEval – 5.87, Codeforces-A – 15.22), arguing intrinsic supervision interferes where an external verifier already exists.

Real-Time Job Ingestion and Deduplication Platform | *Python, GitHub Actions, Concurrent I/O, REST APIs, JavaScript*

- Built a concurrent data-ingestion service in Python polling 950+ external REST APIs across 6 platforms, sweeping 52K+ records per run with a bounded thread pool and per-source fault isolation.
- Deployed on an hourly CI/CD schedule with tiered polling, byte-stable output to avoid churn commits, and write guards that refuse to overwrite good data when upstream sources degrade.

Financial Document Intelligence: Production RAG System | *RAG, GPT-4o, FAISS, BM25 Hybrid Search, Python, Flask, Docker, AWS EC2*

- Designed a production RAG pipeline over 100+ financial prospectuses using dense FAISS embeddings with BM25 hybrid re-ranking and GPT-4o generation, improving retrieval precision by 32% over a TF-IDF baseline.
- Deployed low-latency REST endpoints with per-answer source attribution and confidence scoring, reducing analyst document retrieval time by 50%.

Multimodal Medical AI: Chest X-Ray and Clinical Notes | *ViT-B/16, ClinicalBERT, PyTorch, Cross-Attention, MIMIC-III*

- Developed a multimodal disease prediction system fusing chest X-ray features (ViT-B/16) with clinical note embeddings (ClinicalBERT) via cross-attention, achieving $F1 = 0.89$ on MIMIC-III and outperforming unimodal baselines by 14%.
- Ran ablations across early, late, and cross-attention fusion; applied class-weighted loss and threshold calibration for MIMIC-III label imbalance, reducing false negatives on critical findings by 18%.

TECHNICAL SKILLS

Languages: Python, C++, Java, JavaScript (Node.js), R, SQL, CUDA (basic)

AI / ML: PyTorch, TensorFlow, HuggingFace Transformers, JAX (basic), LoRA / PEFT, DeepSpeed ZeRO, Scikit-learn, XGBoost, OpenCV

LLM & Agents: Fine-tuning (SFT, LoRA, QLoRA), Prompt Engineering, Chain-of-Thought, RAG, Tool-use, Agentic Pipelines, AWS Bedrock, OpenAI API, Anthropic API

Evaluation & Alignment: Self-Correction Analysis, Per-instance Correction Matrix, Ablation Design, Reward Modeling, ROUGE, BERTScore, Hallucination Detection

MLOps: AWS (EC2, S3, Bedrock, SageMaker), Docker, Git, FAISS, SLURM, Weights & Biases, Experiment Tracking, REST APIs

Research Areas: LLM Alignment, Foundation Models, Multimodal Learning, Efficient Inference (Quantization, LoRA), Agentic AI, NLP

ACHIEVEMENTS & PUBLICATIONS

- **Second Thoughts: When LLM Self-Correction Helps vs Hurts** – Course paper (CSCE 638, NLP), Texas A&M; 96-condition study across 71,595 pairs, novel D1/D2 SFT datasets with DeepSeek-R1 attribution-conditioned LoRA fine-tuning, +17.82 GSM8K gain on Mistral-7B.
- **AI/ML Club Sub-Lead**, IIIT Kottayam (2024) – Directed 8 ML workshops and 3 hackathons; grew active membership by 40% and mentored 30+ students. **IEEE Event Organizer** (2023) – Hosted the IoT Cloud Research Symposium with 200+ attendees across 6 research sessions.
- **Hackathon Winner** – Best Idea Award, Shipy Hackathon 2025 (40+ teams) for an autonomous standup-to-Jira pipeline; National Finalist, Smart India Hackathon competing against 500K+ participants across institute, regional, and national rounds.
- **Findings of EMNLP 2026 (Accepted)** – *Deep Learning Job Runtime Prediction for GPU Cluster Scheduling*; co-authored survey on deep learning job runtime prediction for GPU cluster scheduling, in collaboration with XPerf and Texas A&M University, grounded in a 941-job, 37-feature benchmark across 9 GPU configurations plus a 1,884-run inference dataset.